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Does Floor Trading Matter?

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ABSTRACT

Although algorithmic trading now dominates financial markets, some exchanges continue to use human floor traders. On March 23, 2020 the NYSE suspended floor trading because of COVID-19. Using a difference-in-differences analysis around the closure of the floor, we find that floor traders are important contributors to market quality. The suspension of floor trading leads to higher spreads and larger pricing errors for treated stocks relative to control stocks. To explore the mechanism, we exploit two partial floor reopenings that have different characteristics. Our finding suggests that in-person human interaction facilitates the transfer of valuable information that algorithms lack.

SINCE ITS INCEPTION IN 1792, the New York Stock Exchange (NYSE) has relied on floor traders—human beings who trade securities while standing on the floor of the exchange. Over the last 20 years, floor traders have been increasingly replaced by electronic algorithms. Proponents of electronic trading argue that human interaction is unnecessary and electronic algorithms

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are more efficient and capable.[1] Others argue that human traders provide intuition and knowledge that algorithms cannot offer.[2] Empirically, however, whether floor traders improve or harm liquidity and price efficiency in modern markets remains an open question.

We find that human floor traders improve market quality. On March 23, 2020, the NYSE suspended all floor trading and moved to fully electronic trading in response to the COVID-19 pandemic. Using the abrupt change as an exogenous shock to floor trading activity, we conduct a difference-in-differences analysis to identify the effect of floor traders. We find that market quality deteriorates after floor traders are removed. In particular, across a wide variety of specifications, we find that the removal of floor traders leads to higher proportional effective spreads and larger pricing errors. Moreover, the effects occur during continuous trading sessions as well as during the opening and closing auctions. Because the NYSE eventually reopened the floor in phases with different properties, we are able to test for the underlying mechanism.[3] We find evidence of two, nonexclusive, mechanisms: (i) the floor facilitates the transfer of information between designed market makers (DMMs) and floor brokers in a way that electronic trading cannot replicate, and (ii) the use of special order types that only floor traders can access allows floor traders to improve auction outcomes. Our results show that floor trading does matter, even in the age of electronic trading.

Given the increasingly popular belief that "artificial intelligence (AI) will disrupt labor markets" (Grennan and Michaely (2020)) and the increasingly large role played by algorithms, many have argued that floor traders are no longer necessary. The mere fact that humans are being replaced by algorithms suggests that the algorithms have advantages along certain dimensions. Whether this change is beneficial or harmful to market quality is another question, and it is important to understand how these recent changes affect market quality (O'Hara (2015)). As of March 2020, there are 13 registered equity exchanges in the United States. All except the NYSE are 100% electronic—only the NYSE has continued to use human floor traders.[4]

One major challenge in examining the role of floor traders is that their behavior is not exogenous from stock characteristics. Buy-side investors decide

[1] Hu and Murphy (2020) argue that NYSE closing auction quality is worse than NASDAQ closing auction quality, which they attribute to the existence of NYSE floor traders. Similarly, many press articles argue that floor traders cannot compete with algorithms (e.g., Ingram (2003), Kaplan (2020)). Elstein (2005) declares there will be "a quick demise for the floor once alternative is available."

[2] See, for example, Condon and Babwin (2015), Detrixhe (2017), and Byrne (2019).

[3] As we discuss in Section III, the floor partially reopened in three phases on May 26, 2020, June 17, 2020, and May 10, 2021. We use the different characteristics of these three events to help distinguish the economic mechanism that relates floor trading to market quality.

[4] While the NYSE is the only U.S. equity exchange to continue using floor traders, floor traders are still in use on the Chicago Board Options Exchange and the Chicago Mercantile Exchange, among others. Moreover, floor trades are an important component of the NYSE. According to the NYSE (2020b), D orders (which are floor trades only) accounted for 32.9% of closing auction orders in January, 34.5% in February, and 27.6% in March 2020.

whether to use NYSE floor brokers to carry out their orders, and their choices are likely related to stock-level measures of market quality. As such, a simple regression that examines the relation between market quality and floor trader activity might be biased. We address this concern by developing two identification strategies. Both research designs take advantage of the suspension of floor trading activity as an exogenous shock.

Under the first approach, we employ a difference-in-differences analysis that examines market quality for NYSE-listed stocks before versus after the suspension of floor trading and compares these changes to those of a matched sample of NASDAQ stocks that do not experience a change in floor trading activity. Under the second approach, we employ a difference-in-differences analysis that compares trading in NYSE stocks on the NYSE to trading *in these same stocks* on other exchanges during the same time period. For example, the first approach compares market quality in IBM, which is listed on the NYSE, to market quality in a stock listed on NASDAQ that is matched on price, trading volume, market capitalization, and industry. The second approach compares market quality for IBM trades executed on the NYSE to market quality for IBM trades executed on other trading venues during the same period. Both approaches produce economically similar results.

Our identification approach requires the standard parallel trends assumption.[5] It assumes that the treatment group would have evolved in a similar fashion as the control group if floor trading had not been suspended. The difference-in-differences regressions include firm and time fixed effects, which absorb time-invariant firm characteristics and time-varying aggregate shocks. However, it remains possible that time-varying firm characteristics could be different between the treatment and control groups. For example, when we use NASDAQ stocks as a control group, it is possible that NASDAQ stocks are differentially affected by the economic impact of COVID-19 relative to NYSE stocks. We address this concern in several ways.

First, Figures 1 and 2 show that the parallel trends assumption finds support. Visual evidence suggests that both control groups evolve similarly to the treatment group prior to the announcement and suspension of floor trading. Because financial markets start reacting to COVID-19 in late February before the suspension of floor trading, this suggests that NASDAQ stocks were not differentially affected. Under our second identification approach, we compare trading in NYSE stocks on the NYSE to trading in those same stocks on other exchanges during the same time period. This approach allows us to include firm $\times$ date fixed effects. The specification accounts for time-varying firm-level shocks. Thus, a confounding omitted variable, if it exists, would need to differentially affect trading on one exchange, relative to another exchange, in the same stock starting around March 23, 2020. To the best of our knowledge, the

[5] Importantly, treatment is not staggered in our setting, so the issues raised in Goodman-Bacon (2021), and reiterated in a finance setting in Baker, Larcker, and Wang (2022), are not a concern in our setting.

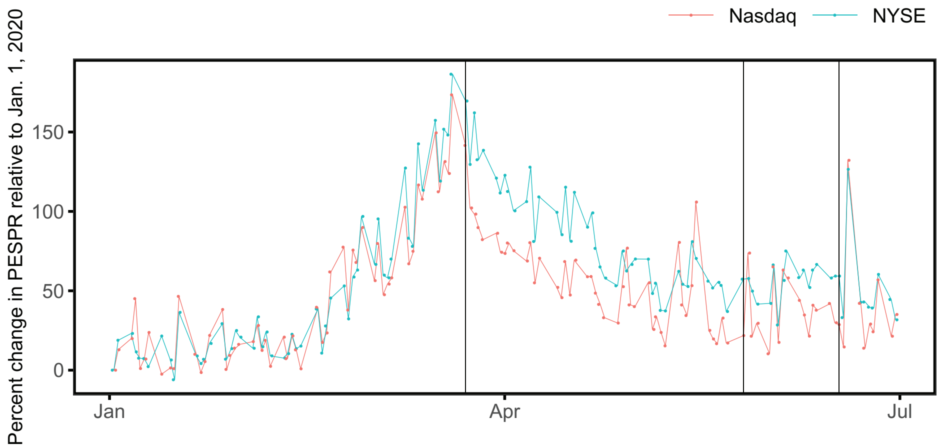

Figure 1. Percent change in daily PESPR for NYSE stocks relative to NASDAQ stocks. This figure shows the change in proportional effective spreads (*PESPR*) for all matched NYSE and NASDAQ-listed stocks in our sample. Each date, for each stock, we calculate the percent change in *PESPR* relative to January 1, 2020 and plot the daily market cap-weighted average *PESPR* across all stocks in our sample. *PESPR* is estimated as the trade price across all exchanges in excess of the prevailing midpoint price using NBBO bid and ask prices. Vertical lines indicate the closing (March 23, 2020), first partial reopening (May 26, 2020), and second partial reopening (June 17, 2020) of the NYSE trading floor. (Color figure can be viewed at wileyonlinelibrary.com)

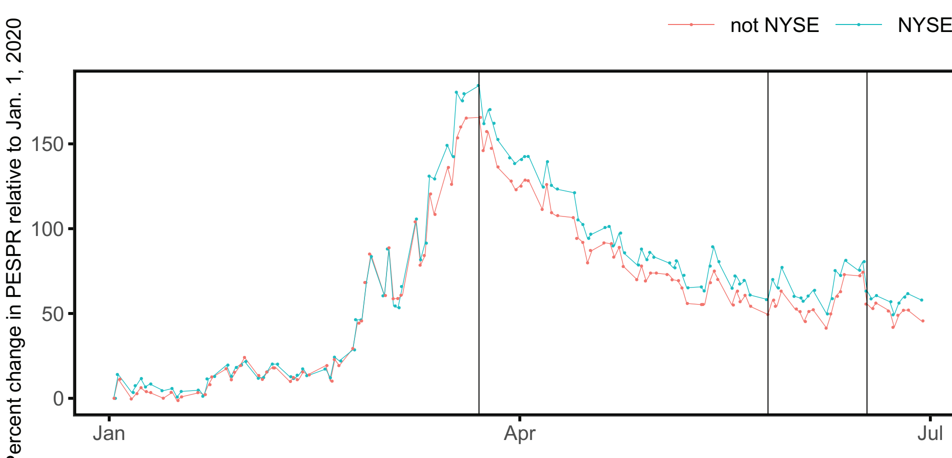

Figure 2. Percent change in daily PESPR for NYSE-listed stocks traded on the NYSE versus off the NYSE. This figure shows proportional effective spreads (*PESPR*) calculated from trades on the NYSE and all other exchanges (not NYSE). Each date, for each stock, we calculate the percent change in *PESPR* relative to January 1, 2020 and plot the daily market cap-weighted average *PESPR* across all stocks in our sample. *PESPR* on the NYSE is estimated from trade prices on the NYSE in excess of the prevailing midpoint price using bid and ask prices on the NYSE. *PESPR* on other exchanges is estimated from trade prices off the NYSE in excess of the prevailing midpoint price using NBBO without NYSE bid and ask prices. Vertical lines indicate the closing (March 23, 2020), first partial reopening (May 26, 2020), and second partial reopening (June 17, 2020) of the NYSE trading floor. (Color figure can be viewed at wileyonlinelibrary.com)

only major change to affect one exchange but not other exchanges around this period is the suspension of floor trading on the NYSE.

We start by examining liquidity. Specifically, we examine proportional effective spreads using the two identification strategies discussed above. Under both identification approaches we find a significant increase in proportional effective spreads for treated stocks relative to control stocks after the suspension of floor trading. For example, proportional effective spreads for trades on the NYSE versus trades in the same stock on other exchanges experience an 11% increase. Moreover, the coefficients are stable as we add fixed effects and controls, which supports our identification assumptions.

To reduce potential confounding effects, we focus the analysis on a short window around the closure of floor trading using data from March 16, 2020 to March 27, 2020. However, it is possible that the effects we document are driven by the sudden and frantic nature of the shift away from floor trading. More specifically, it is possible that once market participants adapt to the new market structure, the effects we document would evaporate or even go in the other direction. However, we find that the effects persist for several months and do not reverse until the floor is reopened.

We next examine price efficiency to see if prices evolve differently in the absence of floor traders. We find that they do. Specifically, we use the Hasbrouck (1993) pricing error measure, which decomposes stock prices into an efficient component and an error component; higher values of the pricing error indicate worse price efficiency. We find that pricing errors increase, that is, the price process becomes noisier for stocks that lose their floor traders. Under our first identification strategy, which uses NASDAQ stocks as a control group, we find that pricing errors for NYSE stocks increase approximately 6% relative to the control group after floor trading is suspended. Under our second identification strategy, which compares trading in the same stock on two different exchanges, we find that pricing errors increase approximately 2% after floor trading is suspended. Overall, the results consistently show that market quality deteriorates after the removal of floor traders.

We also examine when the effects occur within the trading day. Trading volume is highest at market open and market close, and lowest around the middle of the day (Wood, McInish, and Ord (1985)). Moreover, firm-specific information is often released overnight (e.g., Dong, Polk, and Skouras (2019)), which suggests that trading at the beginning of the day is crucial to impounding information. Accordingly, we partition the continuous trading session into half-hour intervals to see if floor traders matter more when trading intensity is highest. We find that they do. Specifically, the results show that floor traders are most important in the hours immediately after the opening auction, with some evidence that they also matter more prior to the closing auction. These findings suggest that floor traders matter more when trading intensity is highest.

We next examine the opening and closing auctions, which also tend to be periods of high trading intensity. Because these auctions have unique market structures, it is difficult to examine them using traditional measures of market quality. Instead, we construct a measure of auction price deviation following

Bogousslavsky and Muravyev (2019). The measure compares the absolute value of the difference between the auction price and a benchmark price. Higher values indicate worse auction quality. Across a variety of specifications for both opening and closing auctions, we find that price deviations get worse after the floor closed.

Our results thus far consistently show that the removal of floor traders is associated with worse market quality, both in continuous trading sessions and during auctions. A natural question to arise is why. The suspension of floor trading halted all floor trading activity. The NYSE operates a hybrid system that combines electronic and floor trading. Traditionally, human traders are physically present on the floor of the exchange and consist of DMMs and floor brokers (collectively, "floor traders"). DMMs are selected by the exchange to provide liquidity, while floor brokers are employees of member firms who trade on behalf of their clients. Importantly, both DMMs and floor brokers could continue trading electronically after the floor closed. Thus, the closure of the floor effectively led to three changes: (i) DMMs and floor brokers could no longer interact *in person*, (ii) the opening and closing auctions for each stock could no longer be conducted manually, and (iii) floor traders could no longer use special orders, in particular, D orders and verbal interest orders.[6]

These three changes offer three possible mechanisms for our findings. The reopening of the NYSE floor provides a way to disentangle them, as the NYSE reopened the floor slowly, in phases that had different characteristics. The first partial reopening occurred on May 26, 2020, when floor brokers but not DMMs returned to the exchange floor. This partial reopening allowed floor brokers to resume using special orders (in particular, D orders), but it did not allow for in-person communication between floor brokers and DMMs (since DMMs did not return to the floor), nor did it allow DMMs to manually conduct auctions (Securities and Exchange Commission (2022a)). A second partial reopening occurred on June 17, 2020 when DMMs returned to the floor (Securities and Exchange Commission (2022b)).[7] This allowed for human interaction between DMMs and floor traders and allowed DMMs to resume manually operating the auctions.

We apply our difference-in-differences design around these two re-openings to test between the three (nonmutually exclusive) explanations for our results. When we examine continuous trading measures such as spreads, we find no evidence of a reversal after the first reopening, when floor trading privileges like D orders resumed, but we find a strong reversal after the second reopening, when DMMs and floor brokers resumed in-person interaction. Specifically, proportional effective spreads strongly reverse direction only after the second

[6] The closing of the floor also meant that pegged orders and orders with a yielding modifier could not be used (Securities and Exchange Commission (2022a)), and impacted the ability of floor traders to get parity allocations.

[7] While the second reopening allowed floor brokers and DMMs to interact in person, social distancing measures remained in place and headcount remained limited until the floor fully reopened on May 10, 2021.

reopening. When viewed together with our results on trading intensity and information arrival, the findings show that the physical trading floor improves market quality during continuous trading sessions by facilitating the flow of information between floor brokers and DMMs. These results are consistent with the empirical evidence in Battalio, Ellul, and Jennings (2007) and the theoretical evidence in Benveniste, Marcus, and Wilhelm (1992) that personal relationships can lead to lower informational asymmetries that improve market quality. Interestingly, when we examine our measures of auction quality, we find a reversal around the first reopening but not the second reopening. This suggests that D orders also play a role, but only during the auction process.

In sum, our findings provide strong evidence that the closure of the NYSE led to worse market quality across a variety of measures including liquidity, price efficiency, and auction quality. These effects appear to be driven by two (nonmutually exclusive) mechanisms. During continuous trading sessions, the floor facilitates the transfer of information between market participants in a way that electronic markets cannot replicate. Morover, during the opening and closing auction, the use of D orders (which give floor traders an advantage over electronic traders) leads to better auction quality. Importantly, we note that even though D orders provide a trading advantage to floor traders, this does not mean that they would necessarily lead to better auction quality. As such, our results provide important and novel evidence that D orders are associated with better outcomes.

Overall, our paper contributes to the broad literature on the role of human interaction in financial markets. Hendershott and Madhavan (2015) and Barclay, Hendershott, and Kotz (2006) focus on the choice of automation versus human interaction in the bond market. They find that in the bond market there are instances when it is important for humans to be involved. Costello, Down, and Mehta (2020) examine whether human beings can improve upon credit-scoring algorithms when making lending decisions. They find that human discretion can enhance loan decisions. In contrast, Jansen, Nguyen, and Shams (2020) find that loans made by computer algorithms perform better than loans made by human underwriters. Chakrabarty and Moulton (2012) show that stocks have worse liquidity when market makers are attention-constrained, but this effect is smaller after the adoption of automation technology.

Many studies analyze the introduction of a new technology, the most important being the rise of algorithmic and high-frequency trading (Hendershott, Jones, and Menkveld (2011), Brogaard, Hendershott, and Riordan (2014), Chaboud et al. (2014), Shkilko and Sokolov (2020)). We add to this literature by focusing on the exogenous removal of arguably the most influential technology—the human floor trader. We also contribute to the literature that studies the unique aspects of the NYSE. One well-known and well-studied feature of the NYSE is that of the DMM (e.g., Venkataraman and Waisburd (2007), Clark-Joseph, Ye, and Zi (2017), Bessembinder, Hao, and Zheng (2020)). For example, Bessembinder, Hao, and Zheng (2020) find that DMM participation leads to better market quality. Clark-Joseph, Ye, and Zi (2017) similarly find that liquidity got worse when the NYSE was closed and DMMs were removed from the market. Importantly, not all floor traders are DMMs, and DMMs continued to operate after the closure of the NYSE trading floor. Thus, we examine a shock that varies floor trading activity but not DMM activity. As such, our results speak to a different question: namely, whether human interaction on the trading floor provides benefits in modern financial markets. More broadly, our findings are consistent with Sofianos and Werner (2000), who show that floor brokers who are not specialists (i.e., not DMMs) are active contributors to market liquidity.

Most recently, Hu and Murphy (2020) examine whether the different structure of the NYSE closing auction, relative to the NASDAQ closing auction, is associated with better or worse market quality. They argue that market quality around the closing auction tends to be worse for NYSE stocks relative to NASDAQ stocks. However, numerous differences in the closing auction on the NYSE besides the presence of floor traders could explain this finding. Although most of their analyses examine the period 2011 to 2018, when they examine the impact of the removal of floor traders in March 2020 they find no evidence that overnight reversals change, suggesting that floor traders do not adversely affect NYSE closing auctions.[8]

Other papers examine stock prices and market quality shortly after the introduction of electronic traders (Benveniste, Marcus, and Wilhelm (1992), Madhavan and Panchapagesan (2000), Venkataraman (2001), Gwilym and Alibo (2003), Handa, Schwartz, and Tiwari (2004), Jain (2005), Battalio, Ellul, and Jennings (2007)). For example, Jain (2005) shows that stock prices rise and equity risk premiums fall after the introduction of electronic trading in 120 countries around the world. Similarly, Venkataraman (2001) examines liquidity on the NYSE versus liquidity on the Paris Bourse in 1997 and finds that the NYSE structure, which uses a mix of electronic and floor traders, appears to have advantages over the Paris Bourse structure, which is fully automated.

The aforementioned literature pre-dates the implementation of Regulation National Market System (NMS) in 2007 and the significant increase in algorithmic trading that has occurred in recent years. As such, whether floor traders matter in modern equity markets remains an open question. To the best of our knowledge, this is the first paper to compare market quality before versus after the removal of floor traders on an equity exchange. We also add to this literature by developing a novel identification strategy and using it to distinguish between various economic mechanisms.

The rest of the paper proceeds as follows. Section I provides background on the structure of the NYSE and describes both the data and our identification strategy. Section II examines the impact of the floor closure for continuous trading and for the opening and closing auctions. Section III examines the economic mechanism underlying our results and discusses caveats as well as directions for possible future research. Section IV concludes.

[8] As Hu and Murphy (2020) state, "the lack of a complete reversal during the reopening period suggests that the larger dislocations may be unrelated to the floor closure."

I. Background and Methodology

To examine the impact of floor traders on NYSE market quality, we combine data from the Center for Research in Security Prices (CRSP) with the NYSE Trade and Quotes (TAQ) database. Using these data, we examine the relation between floor trading and various measures of market quality in a difference-in-differences framework.

A. Structure of NYSE Trading

The NYSE operates a hybrid system that combines electronic and human trading.[9] Traditionally, human traders are physically present on the floor of the exchange. These "floor traders" consist of DMMs as well as floor brokers. DMMs, who replaced specialists on the NYSE in 2008, are traders selected by the exchange to supply liquidity (Bessembinder, Hao, and Zheng (2020)), while floor brokers are "employees of member firms who execute trades on the exchange floor on behalf of the firm's clients" (NYSE (2022d)). Traders on the NYSE floor operate via an open outcry method whereby traders verbally (and potentially nonverbally) signal information to each other, and they use handheld electronic devices that allow them to manually enter orders as well as use algorithms (NYSE (2022a)).

Each trading day, the NYSE starts with an opening auction. The process starts at 6:30 am eastern time (ET), when orders may be submitted (or cancelled) until the DMM opens trading. Starting at 8:00 am, information about order imbalances is disseminated (up to one time per second). DMMs can open trading starting at 9:30 am. If the security's open price is within 10% of the reference price,[10] the DMM can choose to open trading algorithmically; otherwise, the open must be done manually (NYSE (2021)). After the opening auction, trading proceeds continuously until the closing auction at the end of the day. The closing auction process begins at 3:50 pm ET, by which time all limit on close (LOC) and market on close (MOC) orders must be submitted. After 3:50 pm, information about order imbalances is disseminated (up to one time per second) until close. LOC and MOC orders may be submitted only if they have the effect of reducing order imbalances, and they can only be cancelled in cases of "legitimate errors" (NYSE (2021)). The exception is *discretionary orders* ("D orders"), which are available only to floor brokers and have several special features.[11] D orders may be entered after 3:50 pm regardless of whether they reduce or increase order imbalances and they may be cancelled, modified, or submitted until 3:59:50 pm ET (NYSE (2022c)).

[9] The NYSE claims that this model "combines leading technology with human judgment to prioritize price discovery and stability over speed for our listed companies" (NYSE (2022d)).

[10] The reference price is generally the prior day's closing price.

[11] Floor traders also have access to several other unique order types including pegged orders (Securities and Exchange Commission (2022a)), orders with a yielding modifier (an option whereby the order does not follow the typical time-price ordering of quotes) (Securities and Exchange Commission (2022a)), and "verbal interest" orders which are "orders represented orally by a Floor broker at the point of sale" (NYSE (2019)).

Floor traders receive one other notable perquisite. Most U.S. equity trading venues today use price-time priority when matching trades. Under price-time priority, orders are ranked first by which has the best price, and then by the order in which they arrived. However, the NYSE uses "parity/priority" trading (NYSE (2022b)), which first allocates shares to the order that first submitted the best price, and then prorates any remaining shares among all DMMs, floor brokers, and electronic orders that matched the best price. Because the system deemphasizes speed, it benefits human floor traders over fast algorithms, and Battalio, Jennings, and McDonald (2021) find that this system benefits floor brokers at the expense of other traders, who on average receive fills at worse prices. As of 2017, the NYSE also allows parity trading in NASDAQ-listed stocks, although we find that parity trading volume in these stocks is typically low.[12]

B. Floor Closure and Partial Reopenings

On Wednesday, March 18, 2020, the NYSE announced they would suspend floor trading after a trader tested positive for the COVID-19 virus. Traders began planning for the change immediately, and floor trading was officially suspended starting on Monday, March 23. The closure of the floor led to three main changes:

(i) It stopped all open outcry trading. However, both floor brokers and DMMs were allowed to continue trading electronically from remote locations. DMMs could still provide liquidity, but they could no longer interact in person with each other and with floor brokers.
(ii) The opening and closing auctions could no longer be conducted manually (DMMs could still choose to electronically open/close the security or use the exchange-facilitated auction process).[13]
(iii) The closure of the floor prevented floor brokers from using special order types, in particular D orders.[14]

Over the next two years, the floor slowly reopened in phases. First, a subset of floor brokers were allowed to return to the floor on May 26, 2020. As a result, these traders were able to resume the use of D orders, but DMMs and floor brokers were not able to interact in person (since DMMs were not back on the floor) (Securities and Exchange Commission (2022a)). DMMs were allowed to return to the floor on June 17, 2020. At this point, the floor was mostly

[12] See Figure IA.1 in the Internet Appendix. We also find that our results are robust to excluding these stocks from the sample. The Internet Appendix is available in the online version of this article on *The Journal of Finance* website.

[13] Moreover, this also limited some of the information provided by floor traders: "because the floor is unavailable, the pre-opening indications that human DMMs submit from the Trading Floor to the SIP before the open or during trading halts will not be published" (NYSE (2020a)).

[14] The closure also impacted several other special orders including verbal interest orders, pegged orders, and orders with a yielding modifier (Securities and Exchange Commission (2022a)), and it impacted the ability of floor traders to get parity allocation.

back to normal in the sense that D orders could be used and floor brokers and DMMs could interact on the floor, but the headcount remained limited and social distancing measures were still in place (Securities and Exchange Commission (2022b)).[15] Finally, the floor fully reopened on May 10, 2021, when headcount restrictions were lifted.

C. Data

Our sample compares market quality in a narrow window around the announcement and suspension of floor trading, as well as the phased reopening of the floor. In the main analyses examining the floor closure, we use data from March 16, 2020 to March 27, 2020, a two-week window centered around the floor closure. We then examine the reopening of the floor, again using a two-week window centered around two different partial reopenings in May and June 2020. To construct our sample, we start with all 3,447 U.S. common stocks with a CRSP share code of 10 or 11 that are listed on the NYSE (1,256 stocks) or NASDAQ (2,191) in December 2019. We then exclude stocks with more than one share class (431) or with market capitalization below $500 million as of December 2019 (1,412). The resulting sample contains approximately 1,600 equities. From CRSP we get the stock price, dollar trading volume (in thousands of USD), and market capitalization (in millions of USD).

We measure market quality and trading behavior using variables constructed from the TAQ database. We download stock-day proportional quoted spreads (*PQSPR*) and effective spreads (*PESPR*) from the Wharton Research Data Services (WRDS) Intraday Indicators data set. *PQSPR* is the time-weighted average of the difference between the ask and bid prices, scaled by the midpoint price. *PESPR* is the dollar-weighted average of two times the difference between the signed trade price and the prevailing midpoint price, scaled by the midpoint price. See Holden and Jacobsen (2014) for a more detailed discussion on the calculation of these variables.

For some of our analyses, we examine intraday measures calculated using the TAQ database. Because our second identification strategy compares trading in one stock on the NYSE to trading in that same stock on other exchanges, we calculate spreads within each 30-minute interval during the continuous trading session (from 9:30 to 16:00) separately within the NYSE and on all other exchanges. *PQSPR* and *PESPR* on the NYSE are calculated from quotes and trades that occurred on the NYSE. Similarly, *PQSPR* and *PESPR* outside the NYSE are calculated from best bid and ask prices and all trades that occurred off the NYSE.[16] We construct National Best Bid and Offer (NBBO) quotes outside the NYSE following Holden and Jacobsen (2014) and filter trades as in Rösch, Subrahmanyam, and Van Dijk (2017). We also construct

[15] After the second reopening, floor traders were allowed to use all orders types (including D orders) with the exception of verbal interest orders, which were still prohibited due to social distancing measures (NYSE (2020c)).

[16] For example, for this measure, we only use trades on the IEX Exhange (Investors Exchange), NASDAQ, and so on.

Table I
Cross-Sectional Summary Statistics of Time-Series Averages, January 2020 to April 2020

This table reports cross-sectional averages, standard deviations, and 5th, 50th, and 95th percentiles of daily time-series averages by stock. The sample consists of all U.S. common stocks listed on the NYSE or NASDAQ, with a single share class and with a market capitalization of more than USD 500 million as of December 2019. PESPR is the dollar-weighted average of the effective quoted spread (two times the difference between the signed trade price and the prevailing midpoint price, scaled by the midpoint price), Price is the closing price, Volume is trade volume in thousand dollars, Market Cap. is the stock price times shares outstanding in millions of USD in December 2019, and *Log|PricingError|* is the logarithm of the absolute Hasbrouck (1993) pricing error. All variables (except Price and Market Cap.) are estimated over the continuous trading session only, that is, from 9:30 ET to 16:00 ET.

	(1) #Stocks	(2) Mean	(3) *SD*	(4) 5%	(5) Median	(6) 95%
PESPR [%]	1,604	0.18	0.14	0.05	0.14	0.42
Price [USD]	1,604	67.44	104.88	6.90	39.73	207.74
Volume [thousand USD]	1,604	175,799	591,139	3,151	39,796	682,608
Market Cap. [million USD]	1,604	17,193	61,456	623	3,336	67,527
Log|PricingError|	1,599	−8.60	0.77	−9.85	−8.57	−7.44

[Correction added on 18 November 2024, after first online publication: In section I.C, the terms 'millions' and 'million' have been revised to 'thousands' and 'thousand' in three instances in this version.]

Hasbrouck (1993) pricing errors as in Rösch, Subrahmanyam, and Van Dijk (2017). The Hasbrouck (1993) method decomposes prices into an efficient (i.e., random walk) component and a pricing error (i.e., stationary) component. We use the Hasbrouck (1993) pricing errors as a measure of illiquidity. Intuitively, trades that lead to larger price impact will lead to larger pricing errors. Finally, in some of our tests we control for parity trading using the methodology in Battalio, Jennings, and McDonald (2021) and D orders using the methodology in Bacidore, Polidore, and Xu (2014).[17]

Table I contains summary statistics for our sample.

Because our analysis focuses on floor trading in NYSE stocks, the stocks in our sample skew toward larger stocks with relatively high liquidity. The mean (median) stock in our sample has a market capitalization of \$17 billion (\$3.3 billion) and the mean (median) effective spread is 0.18 (0.14) as a percentage of the midpoint price.

D. Identification Strategy

One challenge in identifying the effect of floor traders is that their activity is likely not exogenous from firm-level characteristics. As a result, coefficient estimates from an ordinary least squares (OLS) regression of firm-level characteristics on floor trading activity would likely be biased. Accordingly, to identify

[17] See Section I of the Internet Appendix for a detailed overview of our measure of parity trading. We thank the NYSE for helpful conversations and for providing data to estimate these measures.

the effect of floor traders we use a difference-in-differences regression. To motivate the difference-in-differences regression and formalize our identification assumptions, we first assume that firm-level measures of market quality in the no-treatment state can be expressed according to the additive model

$$E[MarketQuality_{0,i,e,t}|i,e,t] = \rho_e + \lambda_t + \Gamma_i + \alpha_{i,t}, \tag{1}$$

where $MarketQuality_{0,i,e,t}$ is market quality in stock i on exchange e on date t, with the subscript 0 indicating that floor trading is allowed; $MarketQuality_{1,i,e,t}$ is market quality in stock i on exchange e on date t, with the subscript 1 indicating that floor trading is not allowed (i.e., the treatment event occurred). Equation (1) assumes that market quality in the no-treatment state ($MarketQuality_{0,i,e,t}$) is determined by the sum of four variables: (i) ρ_e, the impact of trading on a particular exchange, (ii) λ_t, a time effect that impacts market quality for all firms on a particular date, (iii) Γ_i, a time-invariant firm effect, and (iv) $\alpha_{i,t}$, which captures time-varying firm-level shocks. Equation (1) therefore says that market quality in a stock is determined by where a stock trades (ρ_e), aggregate market conditions (λ_t), time-invariant firm-specific conditions (Γ_i), and time-varying firm conditions ($\alpha_{i,t}$).

For example, consider two stocks, A and B, and assume that stock A is consistently more liquid than stock B. The persistence difference in liquidity is captured by Γ_i while $\alpha_{i,t}$ captures time-varying differences that occur on certain dates (e.g., when stock A has an earnings announcement its liquidity might change). Aggregate shocks that affect both stock A and B (for example, the outbreak of a global pandemic) are captured by λ_t. Finally, ρ_e measures the portion of liquidity that is due to trading on a particular exchange (e.g., NYSE vs. NASDAQ).

In this paper, our goal is to understand whether the presence of floor traders affects market quality (either positively or negatively). Assuming the additive model shown in equation (1), observed market quality can be written as

$$MarketQuality_{i,e,t} = \rho_e + \lambda_t + \Gamma_i + \alpha_{i,t} + \beta \mathbb{1}_{FloorTrading} + \xi_{i,e,t}, \tag{2}$$

where $\mathbb{1}_{FloorTrading}$ is an indicator variable that equals 1 if floor trading is suspended, and 0 otherwise. Our goal is to test whether β is nonzero to see if floor traders affect market quality.

The difference-in-differences estimator compares (the expected value of treated firms *after* treatment minus the expected value of treated firms *before* treatment) minus (the expected value of control firms *after* treatment minus the expected value of control firms *before* treatment). In our setting (i.e., given equation (2)), this yields

$$\begin{aligned} &(\rho_e - \rho_e + \lambda_{t=after} - \lambda_{t=before} + \Gamma_i - \Gamma_i + \alpha_{i,t=after} - \alpha_{i,t=before} + \beta) \\ &-(\rho_e - \rho_e + \lambda_{t=after} - \lambda_{t=before} + \Gamma_j - \Gamma_j + \alpha_{j,t=after} - \alpha_{j,t=before}), \end{aligned} \tag{3}$$

which reduces to

$$= \beta + (\alpha_{i,t=after} - \alpha_{i,t=before}) - (\alpha_{j,t=after} - \alpha_{j,t=before}), \tag{4}$$

where i indexes treatment firms and j indexes control firms. Equation (4) highlights that a traditional difference-in-differences regression will recover the treatment effect from floor trading (β) when the time-varying firm-level shocks in the control group ($\alpha_{j,t}$) are the same as in the treatment group ($\alpha_{i,t}$). This is the traditional parallel trends assumption.

We use two sets of control groups in our difference-in-differences regressions. The first group uses NASDAQ stocks matched on observable characteristics to generate a counterfactual, while the second group uses market quality for NYSE-listed stocks traded *off* the NYSE as a counterfactual for the treatment group. For example, the first group compares market quality in IBM, which is listed on the NYSE, to market quality in a stock listed on NASDAQ that is matched on price, trading volume, market capitalization, and industry. The second approach compares market quality for IBM trades executed on the NYSE to market quality for IBM trades executed on other trading venues during the same time period. Below, we discuss these two approaches in greater detail.

D.1. Approach 1: Matched Sample Comparing NYSE versus NASDAQ Stocks

Our first approach uses a matched sample to form a control group. Our treatment group consists of NYSE-listed stocks, and we use NASDAQ-listed stocks matched on observable characteristics as the control group. As in Rindi and Werner (2019), we match stocks using the following observable variables: price, trading volume, and market capitalization. We also match on industry as measured by the Fama and French 48-industry classification. We add industry as a matching variable to control for the possibility that the pandemic differentially affected stocks in certain industries, which may not be represented equally across exchanges. For example, if the economic shock from the pandemic affected technology stocks differently than blue-chip stocks, a comparison of all NYSE stocks to all NASDAQ stocks could lead to biased estimates as technology stocks are more likely to be listed on NASDAQ.

We use one-to-one nearest-neighbor propensity score matching (PSM) without replacement. Average price and trading volume are both measured one week before the treatment event, from March 16, 2020 to March 20, 2020, while market capitalization is measured as of December 2019.

Table II presents summary statistics for the PSM procedure.

Panel A displays summary statistics for the PSM sample, while Panel B shows the distribution of propensity scores for the treatment and control groups. The distributions are generally similar across the two groups. Panel C contains summary statistics for key variables by treatment and control groups. As expected, there is no statistically significant difference between the treatment and control groups for volume or market capitalization; while there is a slight difference in price, the variable is statistically significant only at the 10% level. We note that the treatment and control groups do exhibit a statistically significant difference for each of the market quality measures we examine. However, the parallel trends assumption only requires that time-varying

Table II

Overview of Propensity Score Matching

This table reports the propensity score matching (PSM) of treated stocks (NYSE-listed stocks) and control stocks (NASDAQ-listed stocks). We match stocks using one-to-one nearest-neighbor PSM without replacement using average price and trading volume from one week before the event (i.e., from March 16, 2020 to March 20, 2020), Market Cap as of December 2019, and industry using the Fama and French 48-industry classification. The event date is the closure of the NYSE floor on March 23, 2020. Panel A reports summary statistics for the matched sample of treatment and control firms. Panel B reports the 1st, 5th, 50th, 95th, and 99th percentiles of estimated propensity scores for the matched sample. Panel C reports univariate comparisons between stock characteristics of the matched sample of treatment and control stocks.

Panel A: Summary Statistics for the Matched Sample

	#Stocks	Mean	*SD*	5%	Median	95%
PESPR	1,236	0.19	0.15	0.06	0.15	0.44
Price	1,236	64.60	105.49	6.56	39.73	184.64
Volume	1,236	171,775	650,543	2,738	33,531	644,952
Market Cap.	1,236	17,261	66,773	615	2,980	67,193
Log $\lvert$*PricingError*$\rvert$	1,124	−8.33	0.75	−9.56	−8.30	−7.21

Panel B: Estimated Propensity Score Distributions

	1%	5%	50%	95%	99%
Treatment	0.36	0.46	0.51	0.55	0.72
Control	0.30	0.43	0.50	0.52	0.57

Panel C: Differences in Covariates Pre-Event

	Treatment	Control	Diff	*t*-Stat
PESPR	0.30	0.44	0.14	7.80
Price	45.02	55.08	10.07	1.96
Volume	240,262	235,010	−5,252	−0.09
Market Cap.	18,955	15,566	−3,388	−0.89
Log $\lvert$*PricingError*$\rvert$	−8.28	−7.91	0.37	8.81

firm-level shocks in the control group are evolving in the same manner as in the treatment group—it does not require that they have the same *level*. As such, the inclusion of firm-fixed effects in our regressions should absorb this level difference.

Using the matched sample, we next examine difference-in-differences regressions of the form

$$y_{i,e,t} = \beta \mathbb{1}_{i,e,t} + \lambda_t + \Gamma_i + \gamma C_{i,e,t} + \epsilon_{i,e,t}, \tag{5}$$

where $y_{i,e,t}$ is a measure of market quality associated with trades in firm i on exchange e on day t. The variable $\mathbb{1}_{i,e,t}$ is an indicator variable that equals 1 if firm i is a NYSE-listed stock (e=treated) and t is after March 23, 2020, and 0 otherwise. λ_t and Γ_i are date and firm fixed effects, respectively. In some specifications, we include stock price and log of dollar trading volume as control variables ($C_{i,e,t}$). We note that if our identification assumptions hold, the addition of these control variables should not change the coefficient on the treatment effect (β) (Oster (2019)). In all analyses, we calculate standard errors clustered by firm.[18]

Of course, matched samples do not necessarily account for unobservable heterogeneity between the treatment and control groups. Our difference-in-differences approach assumes that such unobservable heterogeneity, if it exists, evolves similarly in the treatment and control groups. In our setting, the parallel trends assumption requires that in the absence of treatment, the change in the conditional average market quality of NYSE firms would be equal to the change in the conditional average market quality of the matched NASDAQ firms. There are two primary ways in which this assumption could be violated in our setting. First, if a shock besides the suspension of floor trading occurred around March 23 and differentially effected NASDAQ stocks relative to NYSE stocks, then our treatment estimate might be confounded by this effect. However, to the best of our knowledge, there were no other substantial changes on the NYSE or NASDAQ around this period.[19] Second, if NYSE stocks and NASDAQ stocks were affected differently by COVID-19 and/or the economic effects of COVID-19 then it is possible they would exhibit a different time-varying trend (i.e., $\alpha_{j,t} \neq \alpha_{i,t}$ in equation (2)), in which case NASDAQ stocks would not be a valid counterfactual. Yet, as we discuss in Section II, the treatment effect remains stable across all of our specifications and control groups, suggesting that NASDAQ stocks were affected by COVID-19 in a manner similar to NYSE stocks.

D.2. Approach 2: Within-Stock Variation

Our matched-sample approach uses NASDAQ stocks as a counterfactual for NYSE stocks if there had been no change in floor trading. As discussed above, it remains possible that NASDAQ stocks were differentially affected by COVID-19. To address this concern, we employ a second control group that uses trades for NYSE-listed stocks that occur on other exchanges. The implementation of Regulation NMS in 2007 led to an increase in exchange competition, and hence stocks are now frequently traded on many different exchanges, not just their listing exchange.[20] Our approach compares market quality for IBM stock trades that occur on the NYSE relative to market quality for IBM stock trades that occur on another exchange, before versus after the suspension of floor trading on the NYSE.

[18] Because our main specification does not have a large number of time periods, we do not cluster by time since clustered standard errors are asymptotically consistent as the number of clusters grows large.

[19] We note that COVID-19 caused market disruptions that affected all stocks (λ_t in equation (2)), but this effect is absorbed by time fixed effects in our setting.

[20] See Haslag and Ringgenberg (2023) for a detailed discussion on the effects of Regulation NMS and intermarket competition.

Formally, we examine difference-in-differences regressions of the form

$$y_{i,e,t} = \beta D_{i,e,t} + \kappa_{i,t} + \gamma C_{i,e,t} + \epsilon_{i,e,t}, \tag{6}$$

where $y_{i,e,t}$ measures the market quality associated with trades in firm i on exchange e on day t. The variable $D_{i,e,t}$ is an indicator variable that equals 1 if firm i is a NYSE-listed stock trading on the NYSE and date t is after March 23, 2020, and 0 otherwise. In our main specifications, we include firm $\times$ date fixed effects ($\kappa_{i,t}$) and we calculate standard errors clustered by firm. In some specifications, we include stock price and log of dollar trading volume as control variables ($C_{i,e,t}$). Again we note that if our identification assumptions hold, the addition of these control variables should not significantly change the coefficient on the treatment effect (β).

In our fully saturated specification, we include firm $\times$ date fixed effects. This specification ensures that the treatment effect (β) is estimated using only within-firm variation after accounting for time-varying shocks. Although our matched-sample approach requires the assumption that time-varying shocks to control firms were the same as time-varying shocks to treatment firms ($\alpha_{j,t} = \alpha_{i,t}$), here the firm $\times$ date fixed effects allow us to absorb these time-varying shocks. As a result, the specification accounts for time-varying firm-level heterogeneity. Put differently, time-varying firm-level responses to COVID-19 (or any other time-varying firm effects) are accounted for. A confounding omitted variable, if it exists, would need to differentially affect firm-level trading activity on the NYSE relative to the NYSE *in the same stock*, and it would need to be correlated with the timing of the suspension of floor trading on March 23.

For example, to violate the identification assumption, a confounding variable would have to change market quality in IBM when it trades on NASDAQ relative to market quality in IBM when it trades on the NYSE, starting around March 23, for reasons unrelated to the suspension of floor trading activity. Again, while such an effect is possible, we are not aware of any other changes to the operations of exchanges that occurred on March 23.[21]

D.3. Parallel Trends

Although our identifying assumptions for both approaches are inherently untestable, Figures 1 and 2 provide visual evidence consistent with the parallel trends assumption.

Figure 1 displays the proportional effective spread for trades in NYSE stocks versus trades in NASDAQ stocks. Throughout January, February, and early March, the figure shows that the proportional effective spreads evolved

[21] It is also theoretically possible that the suspension of floor trading on the NYSE caused liquidity providers on other exchanges to change their behavior, which would violate the stable unit treatment value assumption (SUTVA). However, we are not aware of any evidence of this occurring. We also note that the unexpected nature of the change makes it unlikely that other liquidity providers had time to significantly alter their behavior. We discuss this point further in Section III.

in a nearly identical manner for treatment versus control groups. The leftmost vertical line in the figure indicates the floor closure on Monday, March 23. Around this date, the figure shows a dramatic increase in spreads on the NYSE relative to the spreads on NASDAQ. The middle and rightmost vertical lines indicate the first and second partial reopenings of the floor, which occured on May 26, 2020 and June 17, 2020. After the first partial reopening, the gap in spreads between NYSE and NASDAQ stocks appears to remain. The gap appears to disappear after the second reopening, when DMMs returned to the floor and could interact in person with floor brokers.

Figure 2 shows a similar pattern for the proportional effective spread for trades in NYSE stocks versus trades in these same stocks on other venues. The spreads evolve similarly on the NYSE and on other venues until mid-March, at which point the spreads on the NYSE increase significantly relative to spreads off the NYSE. Although a gap between the two spreads seems to appear a few days prior to the official floor closure, we note that the announcement was made the prior week, and social distancing and work-from-home measures were implemented almost instantly. Moreover, we again see that the gap between the two spread measures remains elevated throughout the closure, and largely persists after the first partial reopening. Visually, the figure is suggestive of some (mixed) evidence that the gap narrows after the second partial reopening. Overall, the visual evidence supports our identification assumptions and suggests that the suspension of floor trading led to wider spreads on the NYSE stocks.

II. The Impact of the Floor Closure

In this section, we empirically examine the impact of the floor closure using our two difference-in-differences approaches. In Section II.A, we start by examining whether floor trading matters for market quality during continuous trading (i.e., throughout the trading day, excluding the opening and closing auctions). We find that it does. Across a variety of specifications and control groups, we consistently find evidence that proportional effective spreads and pricing errors deteriorate after the suspension of floor trading. In Section II.B, we examine the impact of the closure on market quality for the opening and closing auctions. We again find that the closure of the floor is associated with worse market quality.

A. Continuous Trading

A.1. Spreads

We start by examining the impact of the floor closure during the continuous trading session using proportional effective spreads (*PESPR*). If electronic liquidity providers are able to successfully replicate, or even improve upon, floor trader activity, then we would expect a zero or negative treatment effect after the removal of floor traders. After the suspension of floor trading, effective spreads should be unchanged or even decrease.[22] If, however, human beings are able to provide some additional skill or facilitate the transfer of information that is not provided by algorithmic liquidity providers, then the suspension of floor trading might lead to an increase in effective spreads.

Table III presents the results. Panel A displays the difference-in-differences treatment effect for a variety of specifications using NASDAQ stocks as a control group. In column (2), which uses stock and date fixed effects, the coefficient of 8.92 on $Treated \times After$ indicates that effective spreads increased significantly after the suspension of floor trading. Although spreads were generally elevated throughout March for both NASDAQ and NYSE stocks as a result of COVID-19, effective spreads on NYSE stocks increased 9 basis points (bps) *more* than spreads on the matched sample of NASDAQ stocks. Relative to the unconditional mean *PESPR* of 12 bps for NYSE trading prior to the closure of the floor, this represents a more than 70% increase in effective spreads as a result of the suspension of floor trading.[23] Moreover, when we control for trading volume and parity trading in columns (3) and (4), the treatment effect is largely unchanged, suggesting that our results are not sensitive to the possibility of omitted variable bias (Oster (2019)).[24]

Panel B displays results from the difference-in-differences analysis using trading in NYSE-listed stocks on other exchanges as a control group. For example, this analysis compares trading in IBM on the NYSE to trading in IBM on other exchanges, like the IEX Exchange and the BZX Exchange, over the same period. The results again show that effective spreads increased as a result of the suspension of floor trading. In our fully saturated specification, shown in column (5), the coefficient of 1.46 on $Treated \times After$ indicates that effective spreads increased 11% for trades on the NYSE, versus trades in the same stock on other exchanges. We note that this specification includes firm $\times$ date fixed effects. As a result, it controls for time-varying firm-level heterogeneity. In addition, as with Panel A, the treatment effect in Panel B is highly stable across specifications. As we vary the fixed effects or add control variables, the treatment effect is largely unchanged, supporting our identification assumptions.[25]

[22] We note that floor brokers might trade with prices closer to midpoint quotes and therefore trades involving floor brokers might have a very small effective spread (Sofianos and Werner (2000)). In a robustness test, we exclude trades at the midpoint and show that our results are robust, though slightly weaker (see Table IA.IV of the Internet Appendix).

[23] According to the NYSE price list (NYSE (2023)), this increase in trading costs is also large relative to the cost of $0.0024 per share for orders that take liquidity via floor broker transactions.

[24] If parity trading were important for market quality in NASDAQ stocks, then the closure of the floor (and subsequent drop in parity trading) could lead to a violation of the SUTVA for our analyses using NASDAQ stocks as a control group. However, as shown in Figure IA.1 in the Internet Appendix, parity trading in NASDAQ stocks tends to be extremely low. Moreover, the point estimates in columns (3) and (4) are not affected by directly controlling for parity trading volume, suggesting that these trades do not impact our conclusions. Finally, we note that our findings are robust to excluding NASDAQ stocks with parity trading.

[25] In Table IA.III of the Internet Appendix, we show that the results are similar when we examine data in a larger window around the floor closure date. We also find that the results are similar for large stocks (Table IA.X) and small stocks (Table IA.XI).

Table III

Difference-in-Differences Regression: PESPR

This table shows results from a difference-in-differences regression of proportional effective spreads (*PESPR*) before versus after the closure of the NYSE floor. Panel A compares NYSE-listed stocks to a matched sample of NASDAQ stocks, while Panel B compares *PESPR* of NYSE-listed stocks trading on the NYSE to *PESPR* in these same stocks trading elsewhere (i.e., off the NYSE). We estimate the following fixed-effect panel regression using data from one week before to one week after the closure of the NYSE floor (March 16, 2020 through March 27, 2020):

$$PESPR_{i,e,t} = \beta_1(Treated_{i,e,t} \times After_{i,e,t}) + \beta_2(Treated_{i,e,t}) + \beta_3(After_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $PESPR_{i,e,t}$ is the effective spread of stock i on day t trading on exchange e, $After_{i,e,t}$ is an indicator variable equal to 1 after the closure of the NYSE floor (March 23, 2020), and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE (Panel A) or if *PESPR* is estimated on the NYSE (Panel B), and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume ($Log(Volume)$) and parity trading volume ($Parityvolume$). In Panel A, *PESPR* is estimated as the trade price across all exchanges in excess of the prevailing midpoint price using NBBO bid and ask prices. In Panel B, *PESPR* on the NYSE is estimated from trades on the NYSE and midpoint prices using bid and ask prices on the NYSE. *PESPR* on other exchanges is estimated from trades off the NYSE and midpoint prices using the NBBO excluding NYSE bid and ask prices (i.e., using the best bid and ask prices across all exchanges excluding the NYSE). t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = *PESPR*			
	(1)	(2)	(3)	(4)
Panel A: NYSE- (*Treated* = 1) versus NASDAQ- (*Treated* = 0) Listed Stocks				
$Treated \times After$	8.92***	8.92***	8.62***	9.20***
	(7.90)	(7.90)	(7.54)	(8.05)
$After$	−11.07***		−9.57***	
	(−11.65)		(−10.26)	
Price			−0.15***	0.18***
			(−4.36)	(5.24)
$Log(Volume)$			4.79***	−3.02**
			(2.89)	(−2.07)
$Parityvolume$			−0.26**	0.13
			(−1.98)	(1.02)
#Stocks	1,236	1,236	1,236	1,236
#Days	10	10	10	10
Within R^2 [%]	1.71	0.59	1.97	0.78
Stock FE	Yes	Yes	Yes	Yes
Day FE	No	Yes	No	Yes

(*Continued*)

Interestingly, the results in this section are consistent with evidence from two recent papers. Using a laboratory experiment, Asparouhova et al. (2020) examine a setting in which human traders have the option to deploy a set of algorithms and to continue trading manually. They theorize that algorithms should lead to better market quality, yet they find that participants who use

Table III—*Continued*

	Dependent Variable = *PESPR*				
	(1)	(2)	(3)	(4)	(5)
Panel B: NYSE Stocks Traded on the NYSE (*Treated* = 1) versus off the NYSE (*Treated* = 0)					
Treated × *After*	1.46***	1.46***	1.46***	1.46***	1.46***
	(2.68)	(2.68)	(2.69)	(2.74)	(3.64)
Treated					12.12***
					(24.92)
After	0.00		−0.00		
	(0.73)		(−0.79)		
Price			−0.00***	0.00	
			(−6.57)	(0.00)	
Log(*Volume*)			−0.04***	−0.05***	
			(−6.93)	(−9.03)	
Within R^2 [%]	0.25	0.11	2.54	1.59	37.56
#Stocks	552	552	552	552	552
#Days	10	10	10	10	10
Exchange × Stock FE	Yes	Yes	Yes	Yes	No
Day FE	No	Yes	No	Yes	No
Day × Stock FE	No	No	No	No	Yes

both algorithms and manual trading perform best. Moreover, they find that algorithmic trading on its own leads to more flash crashes and price volatility. Similarly, Cao et al. (2021) build an artificial intelligence (AI) analyst and show that the AI analyst beats human analysts at forecasting stock prices, but combining the AI analyst with a human analyst leads to the most accurate forecasts.

A.2. Pricing Errors

The previous results show that floor traders matter for liquidity. If decreased liquidity affects trader behavior, it is possible that the removal of floor trading may also affect the price process. Accordingly, in this subsection we examine whether floor traders affect price efficiency. We find that they do. We start by computing the Hasbrouck (1993) measure, which decomposes observed stock prices into an efficient component and a pricing error component; higher values of the pricing error indicate worse price efficiency. After computing the pricing error, we use the log of pricing error as a dependent variable in the difference-in-differences regressions shown in equations (5) and (6). The results are shown in Table IV.

As before, Panel A shows results using the matched sample of NASDAQ stocks as a counterfactual, while Panel B uses trades in NYSE stocks on other exchanges as the control group. In Panel A of Table IV, we find a positive and statistically significant coefficient on *Treated* × *After* in all models. Moreover, the results are again stable across a variety of fixed effects and the inclusion of

Table IV

Difference-in-Differences Regression: Hasbrouck Pricing Error

This table shows results from a difference-in-differences regression of Hasbrouck (1993) pricing errors before versus after the closure of the NYSE floor. Panel A compares NYSE-listed stocks to a matched sample of NASDAQ stocks, while Panel B compares *PESPR* of NYSE-listed stocks trading on the NYSE to *PESPR* in these same stocks trading elsewhere (i.e., off the NYSE). We estimate the following fixed-effect panel regression using data from one week before to one week after the closure of the NYSE floor (March 16, 2020 through March 27, 2020):

$$Log\left|PricingError_{i,e,t}\right| = \beta_1(Treated_{i,e,t} \times After_{i,e,t}) + \beta_2(Treated_{i,e,t}) + \beta_3(After_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $Log\left|PricingError_{i,e,t}\right|$ is the logarithm of the average absolute pricing error of stock i on day t trading on exchange e, $After_{i,e,t}$ is an indicator variable equal to 1 after the closure of the NYSE floor (March 23, 2020), and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE (Panel A) or if the pricing error is estimated on the NYSE (Panel B), and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume ($Log(Volume)$). We estimate Hasbrouck (1993) pricing errors as in section 1.3 of Rösch, Subrahmanyam, and Van Dijk (2017). Using equation (13) in Hasbrouck (1993), we estimate the pricing error associated with each trade, and then average absolute pricing errors across all exchanges (Panel A) and separately by exchange on which the trade occurred (Panel B). t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = $Log\left\|PricingError_{i,e,t}\right\|$			
	(1)	(2)	(3)	(4)
Panel A: NYSE- ($Treated = 1$) versus NASDAQ- ($Treated = 0$) Listed Stocks				
$Treated \times After$	6.28***	6.28***	5.67***	6.48***
	(4.68)	(4.68)	(4.41)	(5.12)
$After$	17.50***		17.10***	
	(17.72)		(17.23)	
$Price$			−0.70***	−0.12
			(−10.03)	(−1.28)
$Log(Volume)$			−9.15***	−19.45***
			(−8.24)	(−16.56)
Within R^2 [%]	11.64	0.39	14.69	4.20
#Stocks	1,122	1,122	1,122	1,122
#Days	10	10	10	10
Stock FE	Yes	Yes	Yes	Yes
Day FE	No	Yes	No	Yes

	Dependent Variable = $Log\left\|PricingError_{i,e,t}\right\|$				
	(1)	(2)	(3)	(4)	(5)
Panel B: NYSE Stocks Traded on the NYSE ($Treated = 1$) versus off the NYSE ($Treated = 0$)					
$Treated \times After$	2.39*	2.39*	2.39*	2.39*	2.39***
	(1.84)	(1.84)	(1.91)	(1.91)	(8.56)
$Treated$					−0.65***
					(−3.43)

(*Continued*)

Table IV—*Continued*

	Dependent Variable = $Log\left\|PricingError_{i,e,t}\right\|$				
	(1)	(2)	(3)	(4)	(5)
Panel B: NYSE Stocks Traded on the NYSE (*Treated* = 1) versus off the NYSE (*Treated* = 0)					
After	22.56***		21.15***		
	(25.18)		(22.43)		
Price			−0.60***	0.09**	
			(−7.40)	(2.13)	
Log(*Volume*)			−10.40***	−20.71***	
			(−9.40)	(−18.01)	
Within R^2 [%]	15.30	0.06	17.91	4.39	1.59
#Stocks	552	552	552	552	552
#Days	10	10	10	10	10
Exchange × Stock FE	Yes	Yes	Yes	Yes	No
Day FE	No	Yes	No	Yes	No
Day × Stock FE	No	No	No	No	Yes

control variables. In columns (1) and (2), the statistically significant coefficient of 6.28 indicates that pricing errors for NYSE stocks increase approximately 6% after floor trading is removed.

In Panel B, when we use trades in NYSE stocks on other exchanges as the control group, we find similar evidence. The coefficient on $Treated \times After$ is positive and statistically significant in all models. The statistically significant coefficient of 2.39 indicates that pricing errors increase approximately 2% during the period when floor trading is removed. In other words, the price process becomes noisier for stocks that lose their floor traders. Overall, the results in Table IV suggest that the suspension of floor trading is associated with an increase in pricing errors. The removal of floor traders leads to worse liquidity as measured by effective spreads, and this change in liquidity leads to worse price efficiency.

A.3. Relation with Trading and Information

It is well-known that trading volume tends to exhibit a U-shape within the day: volume is highest at market open and market close, and lowest around the middle of the day (Wood, McInish, and Ord (1985)). Accordingly, we next partition the continuous trading session into half-hour intervals to see if floor traders matter more when trading intensity is highest. Table V shows effective spreads, by half-hour, using our difference-in-differences analysis with NYSE stocks trading off the NYSE as the control group. Panel A displays results for the morning trading session, while Panel B shows results for afternoon trading.[26]

[26] In Table IA.V of the Internet Appendix, we show that these findings are robust to using our first research design (i.e., comparing NYSE-listed stocks to matched NASDAQ stocks).

Table V

Difference-in-Differences Regression: Intraday PESPR

This table shows results from a difference-in-differences regression of proportional effective spreads (*PESPR*) before versus after the closure of the NYSE floor, comparing *PESPR* of NYSE-listed stocks on the NYSE and off the NYSE. For each 30-minute interval during the day we estimate the following fixed-effect panel regression using data from one week before to one week after the closure of the NYSE floor (March 16, 2020 through March 27, 2020):

$$PESPR_{i,e,t} = \beta_1(Treated_{i,e,t} \times After_{i,e,t}) + \beta_2(Treated_{i,e,t}) + \beta_3(After_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $PESPR_{i,e,t}$ is the proportional effective spread of stock i on day t trading on exchange e, $After_{i,e,t}$ is an indicator variable equal to 1 after the closure of the NYSE floor (March 23, 2020), and 0 otherwise, and $Treated_{i,e,t}$ is an indicator variable equal to 1 if *PESPR* is estimated on the NYSE, and 0 otherwise. *PESPR* on the NYSE is estimated from trade prices on the NYSE in excess of the prevailing midpoint price using bid and ask prices on the NYSE. *PESPR* on other exchanges is estimated from trade prices off the NYSE in excess of the prevailing midpoint price using NBBO excluding NYSE bid and ask prices (i.e., using the best bid and ask prices across all exchanges excluding the NYSE). Panel A displays results during the morning session while Panel B displays results during the afternoon session. In Panel C, we combine all of the half-hour intervals into one panel and interact $After \times Treated$ with a measure of complexity, where complexity is defined as $1 - R^2$ from a regression of returns in stock i on the returns of the market in each half-hour interval. t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

Panel A: Morning	10:00	10:30	11:00	11:30	12:00	12:30
$Treated \times After$	8.16***	5.69***	3.91***	2.39***	−0.13	−0.04
	(5.38)	(6.69)	(4.97)	(3.14)	(−0.20)	(−0.07)
Treated	13.63***	13.39***	12.44***	12.95***	14.15***	12.86***
	(12.56)	(18.45)	(18.32)	(17.84)	(21.49)	(20.73)
Within R^2 [%]	11.10	18.01	23.54	20.21	21.26	18.48
#Stocks	553	553	553	553	553	553
#Days	10	10	10	10	10	10
Stock×Day FE	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Afternoon	13:00	13:30	14:00	14:30	15:00	15:30	16:00
$Treated \times After$	0.63	−0.35	−0.46	−0.86	−0.18	0.00	0.00
	(1.11)	(−0.58)	(−0.74)	(−1.39)	(−0.30)	(−0.52)	(1.53)
Treated	11.99***	11.91***	12.78***	12.35***	12.83***	11.83***	4.33***
	(21.13)	(20.69)	(22.29)	(19.34)	(22.63)	(21.94)	(16.23)
Within R^2 [%]	20.48	18.11	20.02	18.56	22.41	21.30	17.10
#Stocks	553	553	553	553	553	553	553
#Days	10	10	10	10	10	10	10
Stock×Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(*Continued*)

We find that floor traders matter the most immediately following the opening auction when trading volume is high and it is likely that prices are impounding overnight news.[27] Column (1) in Table V shows the treatment effect from the

[27] These findings are consistent with anecdotal evidence. A 2012 *Business Insider* article on the role of floor traders writes that, "Right around the opening bell it's 'complete mayhem'…Then

Table V—*Continued*

Panel C: Explaining *PESPR* by Intraday Variation in Complexity $(1 - R^2)$	
$Complex \times Treated \times After$	17.45***
	(7.33)
$Treated \times After$	−11.94***
	(−7.84)
$Complex \times After$	1.23
	(0.65)
$Complex \times Treated$	22.67***
	(8.51)
$Complex$	58.39***
	(20.44)
Within R^2 [%]	11.13
#Obs.	143,144
Stock×Day FE	Yes

removal of floor trading for the time interval between 9:30 am ET and 10 am ET. The remaining columns show additional half-hour intervals, until markets close at 4 pm ET. Interestingly, the results are strongest in the interval from 9:30 am to 10 am, and they decline monotonically throughout the day. These findings show that human floor traders add the most value in the periods immediately after the opening auction, with weak evidence that they also matter more at the end of the day.[28]

Finally, to explore the relation between floor trading and the arrival of stock specific information, Panel C includes a triple interaction that examines our main treatment effect, $Treated \times After$, multiplied by a measure of stock-specific information complexity. We calculate information complexity as 1 - R^2 from a regression of each stock's return on the contemporaneous market return (as measured by the return on the SPY exchange traded fund) for each half-hour interval. As shown in Morck, Yeung, and Yu (2000), 1 - R^2 measures how much firm-specific information is in a given stock's price. A stock with a low value has returns that are almost perfectly explained by the market's movements, while a stock with a high value has returns that are explained by unique firm-specific information. The positive and statistically significant coefficient of 17.45 on $Complex \times Treated \times After$ in Panel C confirms the main result: floor traders matter more when stock prices are impounding more firm-specific information. These findings are broadly consistent with the predictions in Madhavan and Panchapagesan (2000), who show that designated dealers play a crucial role in price discovery relative to automated trading systems.[29]

around 9:45 a.m. to 10:00 a.m. everything begins to settle down and algos and computer programs do most of the trading" (La Roche (2012)).

[28] Although the last half-hour of the day is not statistically significant at the usual levels, the results are positive at the 15% significance level during this time interval.

[29] The model in Madhavan and Panchapagesan (2000) focuses on the opening auction, not continuous trading. We discuss this finding more in Subsection II.B when we examine opening auction quality.

In our setting, the results show strong evidence the closure of the floor led to worse market quality in times with heavy trading when price discovery is more likely to occur.

B. Opening and Closing Auctions

The results in the previous subsection show that the closure of the floor led to worse market quality during continuous trading sessions. Moreover, the results show that floor traders mattered more during times of day with higher trading volume and more firm-specific information. Accordingly, we next examine whether floor trading matters for the opening and closing auctions. Both the opening and closing auctions are also times when trading volume tends to be heavy and more firm-specific information is impounded into prices. However, both auctions have unique market structures that make it difficult to examine market quality using traditional measures. Accordingly, to examine market quality in these periods we follow Bogousslavsky and Muravyev (2019) and construct a measure of price deviation between the auction price and the midpoint from the NBBO prices in continuous trading. For the opening auction we use the midpoint after 9:30 am and right before the opening auction. For the closing auction we use the midpoint from continuous trading right before the market close. The measure $|Deviation\%|$ is estimated as two times the absolute difference in the logarithm of the auction price and the prevailing (open auction) or last midpoint price during the continuous trading session (close auction) using NBBO bid and ask prices. A higher $|Deviation\%|$ value indicates a lower quality auction. Formally,

$$|Deviation\%| = 2 \times |log(trade) - log(mid)| = 2 \times |log(trade/mid)|, \qquad (7)$$

where $log(trade)$ is log of the auction trade price and $log(mid)$ is the last midpoint price during the continuous trading session. Table VI displays the results using the NYSE versus NASDAQ difference-in-differences setting, with $|Deviation\%|$ as the outcome variable (in basis points).

The first four columns examine deviations around the opening auction, while the last four columns examine deviations around the closing auction. At both the opening auction and the closing auction, when the trading floor closes, the gap between the continuous market price and the auction price widens. Across columns (1) through (4), the results are the same: the coefficient on $Treated \times After$ ranges from a positive and statistically significant 34 bps to 47 bps. Similarly, in columns (5) through (8), the coefficient on $Treated \times After$ is a consistently positive and statistically significant 32 bps, which implies that the floor closure leads to a 100% increase in closing price deviations relative to the unconditional mean.[30] Moreover, this estimate is similar in magnitude and significance even after we include D orders as a control variable in columns (7)

[30] The unconditional mean is estimated across all NYSE stocks in our sample over the two weeks around the floor closure.

Table VI

Difference-in-Difference Regression: Auction Deviation

This table shows results from a difference-in-differences regression of auction price deviations ($|Deviation\%|$) for opening and closing auction prices before versus after the closure of the NYSE floor. We estimate the following fixed-effect panel regression using data from one week before to one week after the event (March 16, 2020 through March 27, 2020):

$$|Deviation\%|_{i,e,t} = \beta_1(Treated_{i,e,t} \times After_{i,e,t}) + \beta_2(Treated_{i,e,t}) + \beta_3(After_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $|Deviation\%|_{i,e,t}$ is the deviation of stock i on day t trading on exchange e, $After_{i,e,t}$ is an indicator variable equal to 1 after the event, and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE, and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume ($Log(Volume)$) and D-orders ($|d - orders|$). $|Deviation\%|$ is estimated as two times the absolute difference in the logarithm of the auction price and the reference price. For the opening auction (columns (1) through (4)) the reference price is the prevailing price, and for the closing auction (columns (5) through (8)) the reference price is the last midpoint price during the continuous trading session using NBBO bid and ask prices. t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = Opening $\lvert Deviation\%\rvert$				Dependent Variable = Closing $\lvert Deviation\%\rvert$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Treated$	34.25*	46.17**	37.38**	46.98***	32.03***	32.03***	32.86***	32.89***
$\times After$	(1.93)	(2.57)	(2.09)	(2.60)	(10.27)	(10.27)	(10.26)	(10.26)
$After$	−110.43***		−129.19***		−16.91***		−15.48***	
	(−6.71)		(−7.32)		(−7.30)		(−6.11)	
$Price$			1.79***	1.07*			0.00	0.04
			(3.20)	(1.74)			(0.02)	(0.39)
$Log(Volume)$			−61.80***	11.04			6.49**	5.68*
			(−3.68)	(0.60)			(2.16)	(1.81)
$\lvert d - orders\rvert$							0.58***	0.56**
							(3.19)	(3.07)
WithinR^2[%]	1.1	0.07	1.27	0.1	1.13	1.14	1.20	1.20
#Stocks	1121	1121	1121	1121	1124	1124	1124	1124
#Days	10	10	10	10	10	10	10	10
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	No	Yes	No	Yes	No	Yes	No	Yes

and (8). The results show that both opening and closing auction quality deteriorate after the suspension of floor trading, and suggest that D orders are not the primary driver of this finding. In sum, all of the results in this section point to the same conclusion: market quality degraded after the floor closed. Spreads and pricing errors increased during continuous trading sessions, and measures of auction quality worsened for both the opening and closing auctions. In the next section, we test for possible economic mechanisms that could be driving these results.

III. The Economic Mechanism

A. Overview of Partial Floor Reopenings

As discussed in Section II, the floor closure led to several simultaneous changes. In particular, (i) DMMs and floor brokers could no longer interact *in person*, (ii) the opening and closing auctions could no longer be conducted manually, and (iii) floor traders could no longer use special order types (especially D orders). Accordingly, our findings in Section II could be due to the lack of in-person interaction, or they could be due to the removal of certain market features (e.g., manual auctions and D orders). To understand the economic forces behind our results, we need to disentangle these competing explanations.

The manner in which the NYSE implemented the partial reopening of the exchange provides an opportunity to test between these mechanisms. As discussed in Section I.B, the NYSE floor reopened slowly in phases. First, on May 26, 2020, a subset of floor brokers were allowed to return to the floor and use special order types, including D orders. However, because DMMs did not return to the floor, they could not manually open and close auctions, nor could they interact with floor brokers in person. Second, on June 17, 2020, DMMs were allowed to return to the floor. This allowed all floor traders to interact in person, and it also allowed DMMs to manually open and close auctions.[31]

Because the first reopening added back D orders, but did not allow DMMs to interact with floor brokers in person, the first reopening provides a natural opportunity to test whether D orders are driving our main findings. Similarly, because the second reopening allowed DMMs to return to the floor, the second reopening allows us to test for the impact of human interaction between floor brokers and DMMs. Moreover, by examining differences between continuous trading and opening/closing auctions, we can also test whether the results are driven by the ability of DMMs to manually conduct auctions.

[31] While the second reopening allowed floor brokers and DMMs to interact in person, social distancing measures remained in place and the headcount remained limited until the floor fully reopened on May 10, 2021. In Table IA.II of the Internet Appendix, we examine the impact of the third and final reopening, and we find that the second partial reopening explains all of the reversal in continuous trading sessions (i.e., the third re-opening has no significant effects). Accordingly, in our main tables we do not include the third reopening.

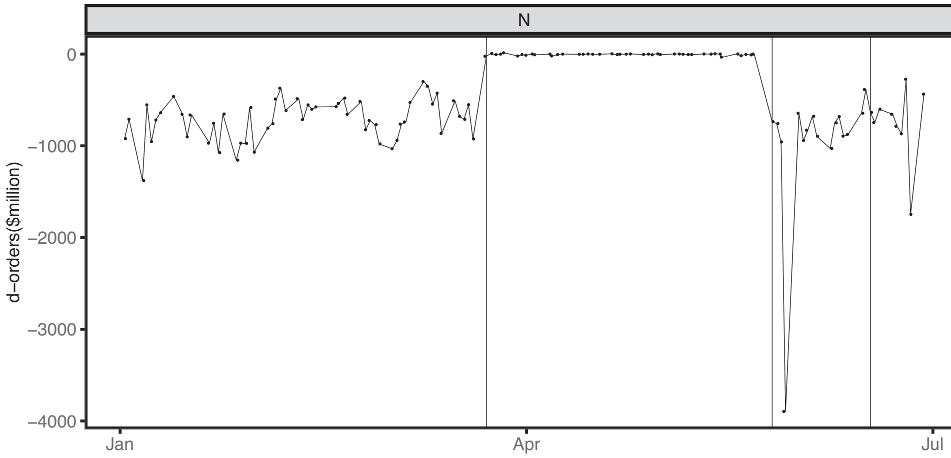

Figure 3. Daily D-orders. This figure shows changes in imbalances at 15:55 in millions of USD across all stocks in the sample. NYSE reports the absolute value and the direction of imbalances every second. At 15:55, D-orders are incorporated into imbalances. Thus, changes in imbalances at 15:55 are a proxy for D-orders, which are only available to floor brokers (Bacidore et al. (2014)). A negative change indicates that imbalances decreased. For each stock i on each day t we estimate the change in imbalances (*ImbalanceQty*) at 15:55 and scale it by the reference price to get a proxy for the volume of D-orders: D-orders $= ReferencePrice \times (ImbalanceQty_{i,t,15:55+} - ImbalanceQty_{i,t,15:55-})$. For each day we then sum all imbalances across all stocks in the given exchange. Vertical lines indicate the closing (March 23, 2020), first partial reopening (May 26, 2020), and second partial reopening (June 17, 2020) of the NYSE trading floor.

B. Effect of Partial Floor Re-openings

To examine the economic mechanism, we apply our difference-in-differences design around the two reopenings. The results for the continuous trading session are shown in Table VII.

In both Panel A (NYSE vs. NASDAQ stocks) and Panel B (within-stock variation), we see similar results—there is no evidence of a reversal in market quality measures at the first reopening. However, in both cases we see a negative and statistically significant coefficient on $Treated \times Open$ 2. Moreover, the results are stable as we add fixed effects and control variables (including parity trading in columns (3) and (4) of Panel A).[32] Thus, spreads do not improve when D orders resume, but they do improve when DMM and floor brokers resume in-person interaction.[33] Importantly, Figure 3 shows that D orders declined to effectively zero when the floor closed, and then rebounded

[32] Because the reopening includes two different events in different periods with different aggregate market conditions, we introduce Stock × Event fixed effects that allow for a different intercept for each reopening. Moreover, in column (5) of Panel B we include Day × Stock × Event fixed effects. The results are stable across all of these specifications.

[33] Moreover, because the first reopening allowed the resumption of other special order types available to floor brokers, this result implies that other floor perquisites including special order types and benefits from parity allocations are not the mechanism driving our findings.

Table VII

Difference-in-Differences Regression around Re-openings: PESPR

This table shows results from a difference-in-differences regression of proportional effective spreads (*PESPR*) before versus after the partial reopenings of the NYSE floor. Panel A compares NYSE-listed stocks to a matched sample of NASDAQ stocks, while Panel B compares *PESPR* of NYSE-listed stocks trading on the NYSE to *PESPR* in the same stocks trading elsewhere (i.e., off the NYSE). We estimate the following fixed-effect panel regression using data from one week before to one week after both times (events) the NYSE Floor partially reopened (May 18, 2020 through June 1, 2020 and June 10, 2020 through June 23, 2020):

$$PESPR_{i,e,t} = \beta_1(Treated_{i,e,t} \times Open1_{i,e,t}) + \beta_2(Treated_{i,e,t} \times Open2_{i,e,t}) + \beta_3(Treated_{i,e,t}) + \beta_4(Open1_{i,e,t}) + \beta_5(Open2_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $PESPR_{i,e,t}$ is the effective spread of stock i on day t trading on exchange e, $Open1_{i,e,t}$ and $Open2_{i,e,t}$ are indicator variables respectively equal to 1 after the first and second partial reopening of the NYSE floor (May 26, 2020 and June 17, 2020), and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE (Panel A) or if *PESPR* is estimated on the NYSE (Panel B), and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume (*Log*(*Volume*)) and parity trading volume (*Parityvolume*). In Panel A, *PESPR* is estimated as the trade price across all exchanges in excess of the prevailing midpoint price using NBBO bid and ask prices. In Panel B, *PESPR* on the NYSE is estimated from trades on the NYSE and midpoint prices using bid and ask prices on the NYSE. *PESPR* on other exchanges is estimated from trades off the NYSE and midpoint prices using the NBBO excluding NYSE bid and ask prices (i.e., using the best bid and ask prices across all exchanges excluding the NYSE). *t*-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = *PESPR*			
	(1)	(2)	(3)	(4)
Panel A: NYSE- (*Treated* = 1) versus NASDAQ- (*Treated* = 0) Listed Stocks				
Treated × *Open*1	0.27	0.27	0.25	0.30
	(0.72)	(0.73)	(0.60)	(0.80)
Treated × *Open*2	−4.85***	−4.85***	−5.35***	−4.93***
	(−6.69)	(−6.69)	(−7.00)	(−6.73)
*Open*1	−0.16		−2.27***	
	(−0.48)		(−5.64)	
*Open*2	5.74***		5.50***	
	(8.72)		(8.45)	
Price			−0.13***	−0.05***
			(−5.38)	(−3.59)
Log(*Volume*)			9.08***	0.06
			(13.42)	(0.13)
Parityvolume			−0.27**	−0.09
			(−2.04)	(−1.09)
Within R^2 [%]	0.76	0.32	5.22	0.33
#Stocks	1,231	1,231	1,231	1,231
#Days	20	20	20	20
Stock × Event FE	Yes	Yes	Yes	Yes
Day FE	No	Yes	No	Yes

(*Continued*)

Table VII—*Continued*

	Dependent Variable = *PESPR*				
	(1)	(2)	(3)	(4)	(5)
Panel B: NYSE Stocks Traded on the NYSE (*Treated* = 1) versus off the NYSE (*Treated* = 0)					
Treated × *Open*1	0.41**	0.41**	0.41**	0.41**	0.41***
	(2.07)	(2.07)	(2.08)	(2.09)	(2.96)
Treated × *Open*2	−1.06***	−1.06***	−1.06***	−1.06***	−1.06***
	(−4.33)	(−4.33)	(−4.35)	(−4.37)	(−8.28)
*Open*1	−0.46***		−0.28**		
	(−4.41)		(−2.32)		
*Open*2	−2.01***		−1.98***		
	(−13.07)		(−12.94)		
Price			−0.01	−0.02	
			(−0.66)	(−0.76)	
Log(*Volume*)			−0.44***	−1.37***	
			(−6.27)	(−12.08)	
Within R^2 [%]	4.82	0.25	5.00	1.32	0.97
#Stocks	550	550	550	550	550
#Days	20	20	20	20	20
Exchange × Stock × Event FE	Yes	Yes	Yes	Yes	
Day FE	No	Yes	No	Yes	
Day × Stock × Event FE	No	No	No	No	Yes

to normal levels after the first reopening with little change after the second reopening. The results therefore provide strong evidence that D orders are not the mechanism behind our results in continuous trading sessions.

The fact that spreads decline after the second reopening, when DMMs and floor brokers both resumed in-person trading, suggest there is something important about human interaction. Indeed, when combined with our results on trading intensity and information arrival, the findings strongly suggest that the physical trading floor improves market quality during continuous trading sessions because it facilitates the flow of information between floor brokers and DMMs. This result is consistent with several other findings in the literature. Battalio, Ellul, and Jennings (2007) examine what happens when specialists change their physical location on the NYSE trading floor, which causes them to interact with a different set of floor brokers. They find that spreads widen until the specialist builds a relationship with these floor brokers.[34] Similarly, Benveniste, Marcus, and Wilhelm (1992) show theoretically that personal relationships between specialists (now called DMMs) and brokers can lead to lower informational asymmetries that improve market quality, and Grammig, Schiereck, and Theissen (2001) find empirically that information asymmetries are lower in floor trading systems relative to computerized trading systems.

[34] See also Koudijs and Voth (2020), who examine market quality in Amsterdam after Jewish market makers were removed from the floor in World War II. They find that market quality degrades and takes five weeks to rebound, which they attribute to the time it takes new traders to learn the necessary skills. We argue that this could also relate to forming new relationships.

Overall, the results suggest that electronic markets are not able to recreate the information transfers that occur with in-person trading. As such, floor traders do improve market quality during continuous trading. But what about the opening and closing auctions? The structure of auctions is quite different from continuous trading. Accordingly, Table VIII examines the two partial reopenings for the opening and closing auctions.

Interestingly, when we examine our measures of auction quality, we find a reversal around the first reopening for both the opening and closing auction. Specifically, we see negative and statistically significant coefficients on $Treated \times Open$ 1, but we see no reversal around the second reopening. Because the first reopening brought back D orders, this suggests that D orders matter, but only during the auction process. This is consistent with recent evidence by Jegadeesh and Wu (2022), who compare the quality of closing auctions on the NYSE to closing auctions on NASDAQ and find that the NYSE closing auction is better because of special privileges accorded to floor trades.

To further explore this result, Table IX again examines the opening auction around the two partial reversals, but uses three different benchmark prices to calculate the measure of auction deviation.

Our main tests calculate opening auction deviation when the benchmark price is the most recent midpoint price *before* the auction. In columns (1) and (2), we instead define the benchmark as the first midpoint price *after* the auction, in columns (3) and (4) as the first midpoint price after 1 pm, and in columns (5) and (6) as the first midpoint price after 4 pm. The results show no significant impact from the reopenings in any of the specifications, but a strong reversal for the second reopening in columns (3) through (6). These results suggest that floor traders traders help make opening prices *closer* to prices at 1 pm and 4 pm (i.e., the distance between these prices is lower when the floor reopens). Moreover, this effect is concentrated around the second partial reopening, which suggests that the opening auction is more efficient when floor brokers and DMMs can communicate in person. The results are consistent with Madhavan and Panchapagesan (2000), who show that designated dealers can improve price discovery at open over a fully automated market.

Overall, our results indicate that the closure of the NYSE led to worse market quality across a variety of measures including liquidity, price efficiency, and auction quality. Moreover, the effects are driven by two (nonmutually exclusive) mechanisms. During continuous trading sessions, the floor facilitates the transfer of information between DMMs and floor brokers in a way that electronic markets cannot replicate. During the opening and closing auctions, the floor facilitates the transfer of information and the use of D orders, which gives floor traders an advantage over electronic traders and leads to better auction quality. In sum, floor trading does matter.

C. Limitations

Although our results provide strong evidence that human floor traders are important contributors to market quality in our setting, we are careful to note

Table VIII
Difference-in-Difference Regression around Reopenings: Auction Deviation

This table shows results from a difference-in-differences regression of auction price deviations ($|Deviation\%|$) for opening and closing auction prices before versus after the partial reopenings of the NYSE floor. We estimate the following fixed-effect panel regression using data from one week before to one week after both times (events) the NYSE floor partially reopened (May 18, 2020 through June 1, 2020 and June 10, 2020 through June 23, 2020):

$$|Deviation\%|_{i,e,t} = \beta_1(Treated_{i,e,t} \times Open1_{i,e,t}) + \beta_2(Treated_{i,e,t} \times Open2_{i,e,t}) + \beta_3(Treated_{i,e,t}) + \beta_4(Open1_{i,e,t}) + \beta_5(Open2_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $|Deviation\%|_{i,e,t}$ is the deviation of stock i on day t trading on exchange e, $Open1_{i,e,t}$ and $Open2_{i,e,t}$ are indicator variables respectively equal to 1 after the first and second partial reopening of the NYSE floor (May 26, 2020 and June 17, 2020), and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE, and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume ($Log(Volume)$) and D-orders ($|d-orders|$). $|Deviation\%|$ is estimated as two times the absolute difference in the logarithm of the auction price and the reference price. For the opening auction (columns (1) through (4)) the reference price is the prevailing price, and for the closing auction (columns (5) through (8)) the reference price is the last midpoint price during the continuous trading session using NBBO bid and ask prices. t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = Opening $\|Deviation\%\|$				Dependent Variable = Closing $\|Deviation\%\|$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Treated \times Open1$	−20.50**	−20.50**	−20.61**	−20.68**	−3.74***	−3.73***	−4.00***	−3.92***
	(−2.45)	(−2.45)	(−2.44)	(−2.46)	(−6.06)	(−6.06)	(−6.50)	(−6.40)
$Treated \times Open2$	37.57***	37.72***	36.99***	37.99***	−0.61	−0.61	−0.97	−0.91
	(6.82)	(6.82)	(6.64)	(6.80)	(−0.89)	(−0.89)	(−1.39)	(−1.32)
$Open1$	25.38***		23.58***		1.47***		−0.12	
	(3.08)		(2.96)		(2.96)		(−0.25)	
$Open2$	−48.81***		−48.95***		2.57***		2.36***	
	(−9.46)		(−9.44)		(4.30)		(4.07)	

(*Continued*)

Table VIII—*Continued*

	Dependent Variable = Opening $\|Deviation\%\|$				Dependent Variable = Closing $\|Deviation\%\|$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Price			−0.24	0.31			−0.09**	−0.10**
			(−0.60)	(0.77)			(−2.09)	(−2.26)
$Log(Volume)$			9.04***	1.51			6.99***	4.89***
			(2.60)	(0.37)			(8.33)	(5.06)
$\|d - orders\|$							0.05***	0.03**
							(3.01)	(2.26)
Within R^2 [%]	0.63	0.19	0.67	0.19	0.39	0.16	3.00	0.99
#Stocks	1,115	1,115	1,115	1,115	1,118	1,118	1,118	1,118
#Days	20	20	20	20	20	20	20	20
Stock × Event FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	No	Yes	No	Yes	No	Yes	No	Yes

Table IX

Difference-in-Differences Regression around Reopenings: Open Auction Deviation with Alternate Reference Prices

This table shows results from a difference-in-differences regression of auction price deviations ($|Deviation\%|$) for opening auction prices before versus after the partial reopenings of the NYSE floor. We estimate the following fixed-effect panel regression using data from one week before to one week after both times (events) the NYSE floor partially reopened (May 18, 2020 through June 1, 2020 and June 10, 2020 through June 23, 2020):

$$|Deviation\%|_{i,e,t} = \beta_1(Treated_{i,e,t} \times Open1_{i,e,t}) + \beta_2(Treated_{i,e,t} \times Open2_{i,e,t})\beta_3(Treated_{i,e,t}) + \beta_4(Open1_{i,e,t}) + \beta_5(Open2_{i,e,t}) + \gamma(C_{i,e,t}) + FE + \epsilon_{i,e,t},$$

where $|Deviation\%|_{i,e,t}$ is the deviation of stock i on day t trading on exchange e, $Open1_{i,e,t}$ and $Open2_{i,e,t}$ are indicator variables respectively equal to 1 after the first and second partial re-opening of the NYSE floor (May 26, 2020 and June 17, 2020), and 0 otherwise, $Treated_{i,e,t}$ is an indicator variable equal to 1 if the stock is listed on the NYSE, and 0 otherwise, and $C_{i,e,t}$ is a vector of additional control variables including the natural log of trading volume ($Log(Volume)$) and D-orders ($|d-orders|$). $|Deviation\%|$ is estimated as two times the absolute difference in the logarithm of the auction price and the reference price. The reference price is the next midpoint price (columns (1) and (2)), the midpoint price at 1 pm (columns (3) and (4)), or the midpoint price at 4 pm (columns (5) and (6)) using NBBO bid and ask prices. t-Statistics calculated using standard errors clustered by stock are shown in parentheses below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent Variable = Opening Auction $\|Deviation\%\|$ using:					
	Mid-Price After Open		Mid-Price at 1 pm		Mid-Price at 4 pm	
	(1)	(2)	(3)	(4)	(5)	(6)
$Treated \times Open1$	4.22	2.68	−3.28	−6.84	12.58	8.25
	(0.15)	(0.10)	(−0.29)	(−0.62)	(0.90)	(0.62)
$Treated \times Open2$	27.36	30.52	−57.75***	−65.03***	−23.43*	−31.35**
	(0.68)	(0.77)	(−4.60)	(−5.24)	(−1.75)	(−2.34)
$Price$		3.23		−1.46***		−1.20
		(0.87)		(−2.14)		(−1.44)
$Log(Volume)$		4.53		172.28***		198.96***
		(0.41)		(15.45)		(14.20)
Within R^2 [%]	0.00	0.02	0.13	3.33	0.02	3.08
#Stocks	1,115	1,115	1,115	1,115	1,115	1,115
#Days	20	20	20	20	20	20
Stock × Event FE	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes

several limitations with our analyses. First, our analyses examine a relatively short time period around the suspension of floor trading. It is possible that algorithms would eventually change their behavior in a way that mitigates the impact of the removal of floor traders. However, Table IA.II of the Internet Appendix examines data using all of 2020 and 2021 and shows that our results are unchanged. Moreover, to further establish the robustness of our main findings, Table IA.II examines our findings for 20 different liquidity measures available in the WRDS Intraday Indicators database.[35] Across these different outcome variables, we consistently find a decrease in market quality when the floor closes and we find no evidence that the adverse impact of the floor closure reverses until the second reopening, in June 2020. The robustness of our findings across variables, specifications, and sample periods suggests that they are unlikely to be a false positive. Nonetheless, it is possible that at even longer horizons the results would differ. Future research should continue to examine the implications of this.

We note that this limitation is related to our identification assumptions. Implicitly, our difference-in-differences regressions assume that other liquidity providers did not significantly change their behavior in response to the suspension of floor trading. If they did, the SUTVA may be violated. In our setting, the rule change was not anticipated and we focus on a narrow window around the suspension of floor trading, making it unlikely that other liquidity providers had time to react. It remains possible, however, that the long-run effects of floor trader removal are different than the effects we document.

It is also possible that buy-side traders chose to trade on different venues as a result of the floor closure because they worried the floor closure would hurt liquidity on the NYSE. In the Internet Appendix, we investigate the effect of floor trading on NYSE market share and find that market share did decrease with the closure of the floor during the continuous trading session and the closing auction, but increased during the opening auction (see Tables IA.VI to IA.IX). However, this effect is still a treatment effect, since it represents an effect stemming from the floor closure. The fact that market share on the NYSE fell as a result of the floor closure is another data point suggesting that floor traders are important for market quality. Without floor traders, the NYSE would attract less volume.

IV. Conclusion

Historically, financial trading has been dominated by human activity. Over the last few decades, however, algorithms have increasingly replaced human traders. As a consequence, the rise of anonymous algorithmic trading raises fundamental questions about financial market structure. In particular, do human traders have a role to play in modern financial markets?

[35] To show robustness to data-mining concerns, this table examines all relevant spread and price impact measures available in the database.

In this paper, we study whether human floor traders have an impact on market quality in U.S. equity markets. We use the suspension of NYSE floor trading on March 23, 2020 as a shock to floor trading activity that is exogenous from firm-level characteristics. The results show that, even in the age of algorithmic trading, floor traders are important contributors to liquidity and price efficiency. Following the suspension of floor trading, we find higher proportional effective spreads and worse pricing errors. We also find worse market quality around the opening and closing auctions.

To understand the economic mechanism underlying these results, we examine two partial reopenings of the floor that each had different characteristics. The first partial reopening allowed floor brokers to resume using D orders, while the second allowed DMMs to resume in-person interaction with floor brokers and conduct manual opening and closing auctions. We find that market quality during continuous trading sessions improves after the second reopening but not the first. In contrast, we find that auction quality improves after the first reopening but not the second. Together, these results suggest that floor traders are important contributors to market quality for two (nonmutually exclusive) reasons: (i) the floor facilitates the transfer of information between DMMs and floor brokers in a way that electronic trading cannot do, and (ii) the use of D orders allows floor traders to improve auction outcomes. Overall, our results show that human floor traders continue to be valuable intermediaries even in the age of algorithmic trading.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.